

Fig. 10: Reactance versus frequency plot in Android

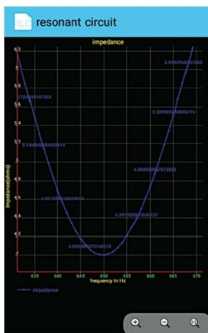


Fig. 12: Impedance versus frequency plot in Android

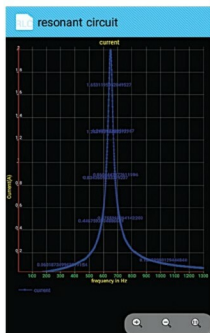


Fig. 14: Current versus frequency plot in Android

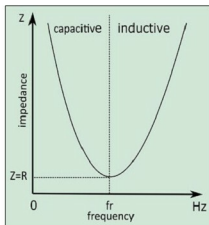


Fig. 11: Impedance versus frequency

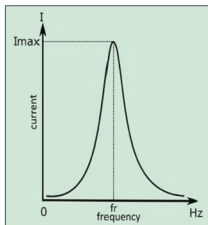


Fig. 13: Ideal case of current versus frequency

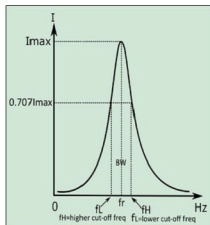


Fig. 15: Ideal case of current versus frequency

frequency and inductor and capacitor now act as a short circuit.

Now, resistor is the only resistive element present in the circuit. So impedance is equal to resistance ($Z=R$). In the graph, below resonance frequency, capacitive reactance is greater than inductive reactance ($X_C > X_L$). And, above resonance frequency, inductive reactance is greater than capacitive reactance ($X_L > X_C$).

In Fig. 10, the blue line indicates the inductive reactance curve and the white line indicates the capacitive reactance curve. When resonance frequency is 649.79Hz, both X_L and

X_C are 81.64 ohms.

Impedance (Z). In Fig. 11, at resonance frequency, inductor and capacitor reactance are equal to and opposite of each other. So these cancel each other out and resistor is the only resistive element present; impedance is equal to the resistance ($Z=R$).

In Figs 11 and 12, impedance gradually decreases as frequency approaches resonance frequency. Where resonance frequency is 649.74Hz, impedance is equal to resistor value $R=40\Omega$.

Series circuit current at resonance. In Fig. 13, current gradu-

ally increases as frequency increases. When frequency approaches resonance frequency, impedance Z of the circuit equals resistance R ($Z=R$), because at resonance condition, the inductor and capacitor act as a short circuit. So, only resistive element is present. Resistor is the only element that opposes the flow of current and at resonance, current is calculated using: $I_{max} = V/R$.

In Fig. 14, current gradually increases as frequency increases. When resonance frequency is 649.7Hz, current becomes maximum, $I_{max}=2A$. This is because, at resonance fre-